



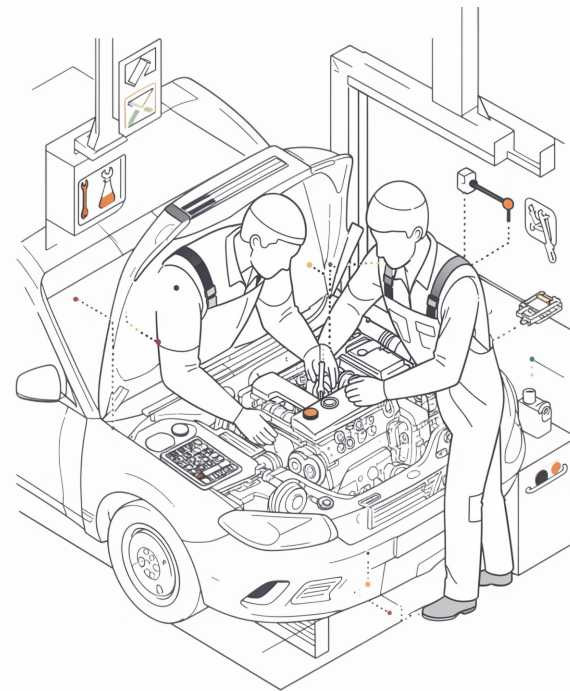
the AI-native operating system and continuous telemetry network that transforms the \$500B independent auto aftermarket from reactive guesswork into a predictive, data-driven science.

Awase Khirni Syed Ph.D.

SycliQ Prognostix OSTM v8.0

AI-Native Operating System for the \$500B Automotive Aftermarket

Shifting auto-repair from reactive guesswork to predictive science.



PrognostixOS: The AI-Native OS for Independent Auto Repair

Transforming the \$500B automotive aftermarket - one prediction at a time.



Current State



Lost Billable Hours

Shops lose **15–20%** of billable hours to fragmented, disconnected tools.



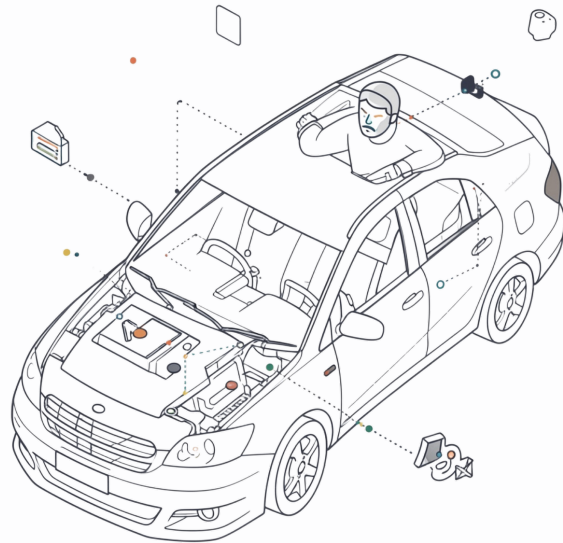
Break-Fix Culture

85% of all repairs are reactive — customers only come in when something breaks.



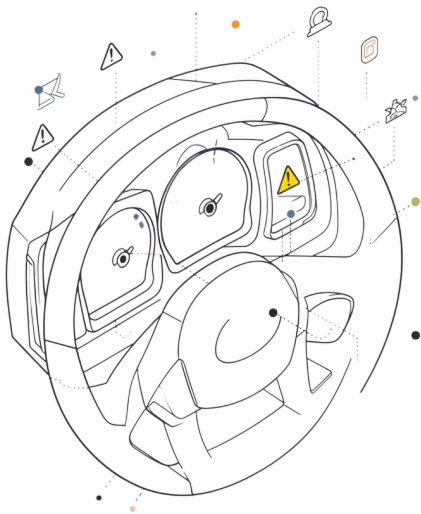
EV Blind Spot

Legacy software **cannot diagnose EV battery degradation** — the fastest-growing segment.





Flying Blind in 2025



The Break-Fix Trap

80% of repairs are reactive. Customers only arrive when a warning light fires – driving anxiety, eroding trust, and leaving revenue on the table.

The Diagnostic Black Box

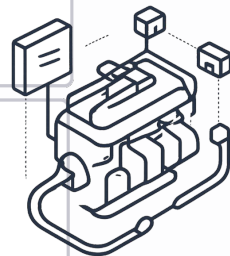
Modern vehicles generate **terabytes of telemetry**, yet shops only see cryptic DTCs. Technicians waste **up to 30% of billable hours** just diagnosing.

The EV & SDV Cliff

Electric Vehicles and Software-Defined Vehicles demand entirely new diagnostic paradigms. **Legacy tools are blind** to cell-level battery health and network topology mapping.

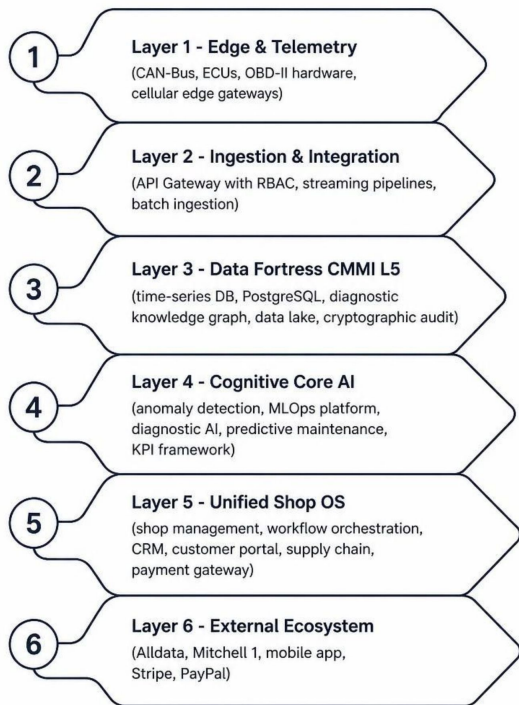
Supply Chain Opacity

Zero real-time visibility into parts pricing, availability, or vendor SLAs. **Bays sit empty** waiting for parts while revenue evaporates.





Six-Layer System Architecture



LAYER 1

Edge & Telemetry

Vehicle CAN-Bus and ECUs feed continuous telemetry through proprietary OBD-II hardware to edge gateways via cellular connectivity.

LAYER 2

Ingestion & Integration

API Gateway enforces rate limits and RBAC, routing data through streaming pipelines for real-time events and batch ingestion for third-party APIs.

LAYER 3

Data Fortress (*CMMI L5 Governance*)

Time-series DB, PostgreSQL, diagnostic knowledge graph, and data lake — governed by cryptographic audit, data quality cleansing, and lineage tracking.

LAYER 4

Cognitive Core (AI & Analytics)

Anomaly detection, MLOps model serving, diagnostic AI, and predictive maintenance — all instrumented via a KPI success framework.

LAYER 5

Unified Shop OS

Shop management, workflow orchestration, CRM, customer portal, supply chain, procurement, and payment gateway integration.

LAYER 6

External Ecosystem

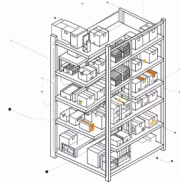
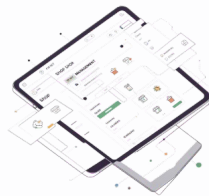
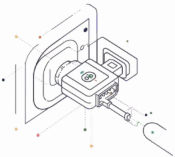
Parts catalogs (Alldata, Mitchell 1), customer mobile app notifications, and payment providers (Stripe, PayPal).



Solution

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SycliQ Prognostix OS Hub



Continuous Telemetry Layer

Proprietary OBD-II hardware captures live CAN-bus and PID data, building a high-fidelity **Digital Twin** of every vehicle.

Cognitive AI Diagnostics

Our **DiagnosticKnowledgeGraph** correlates live telemetry with historical repair data — predicting failures before they happen.

Unified Shop OS

One platform replacing **5+ disjointed tools**: CRM, AI-estimating, workflow management, and integrated payments.

End-to-End Supply Chain

Automated procurement, real-time vendor SLA tracking, and warehouse management integrated directly into repair orders.

Connected Ecosystem

Vehicle Data

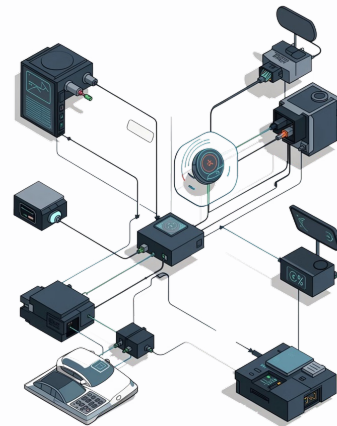
Telemetry & CAN-Bus feeds drive real-time diagnostics at the edge.

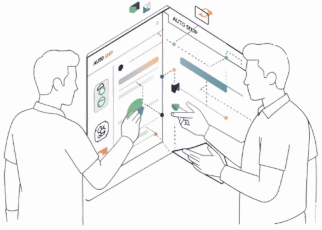
Supply Chain

Parts distributors, payment gateways, and OEM APIs power the parts ecosystem.

Customer Layer

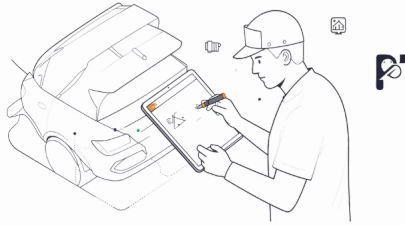
Mobile app, service dashboards, and fleet portals serve every touchpoint.





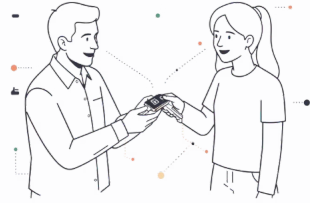
Shop Owners

+25% revenue per bay driven by predictive alerts that bring customers in before failure.



Technicians

40% faster diagnostics with step-by-step AI guidance — less guesswork, more throughput.



Customers

Proactive alerts, transparent estimates, and a service experience that earns **lifetime loyalty**.



Technology Moat

Proprietary Hardware + Data Flywheel

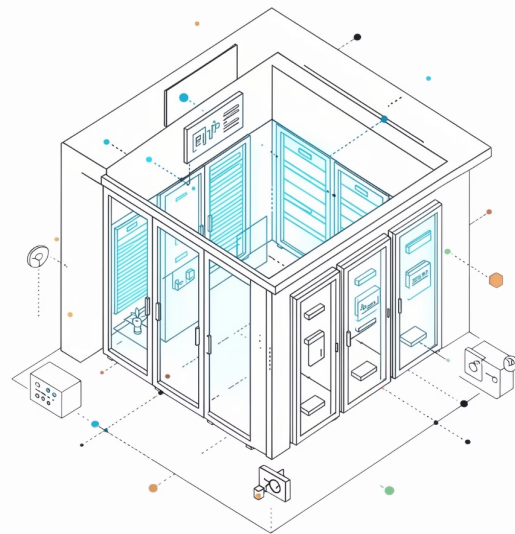
Every mile driven and every repair completed feeds ML models — making predictive accuracy **exponentially stronger** than software-only competitors.

Court-Admissible Data Integrity

CMMI Level 5 compliance, SHA-256 cryptographic audit trails, and automated GDPR classification. Shops win warranty disputes with **cryptographic proof**.

Deep EV & Network Analytics

Native EV battery diagnostics: State of Health, cell deviation, thermal management, and full vehicle network topology mapping.





Solution

Data Flywheel

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Hardware + AI = Exponential Moat

Every mile driven and every repair completed feeds our ML models. Software-only competitors cannot replicate this data advantage.

EV & Network Native

Deep support for battery State of Health, cell deviation, thermal management, and vehicle network topology — the standard no one else is building.

Enterprise-Grade Data Fortress

CMMI Level 5 compliance, cryptographic SHA-256 audit trails, automated GDPR classification. Our data integrity is **court-admissible**.



Shops win warranty disputes, pass fleet audits, guarantee repair quality. Prognostix becomes the trusted standard.



Continuous OBD-II Telemetry + Predictive AI

Real-Time Digital Twin

Proprietary hardware captures **continuous vehicle telemetry**—not just diagnostic codes. Every connected vehicle generates a living, breathing mirror of its mechanical state.

Predictive Power

AI predicts component failures **weeks before they happen**—alternators, batteries, EV cells—turning shops from reactive repair bays into precision maintenance centers.

Automated Supply Chain

Parts ordered **automatically** the moment a failure is predicted; technicians guided step-by-step through repair—zero lag, zero guesswork, maximum throughput.

Enterprise-Grade Security

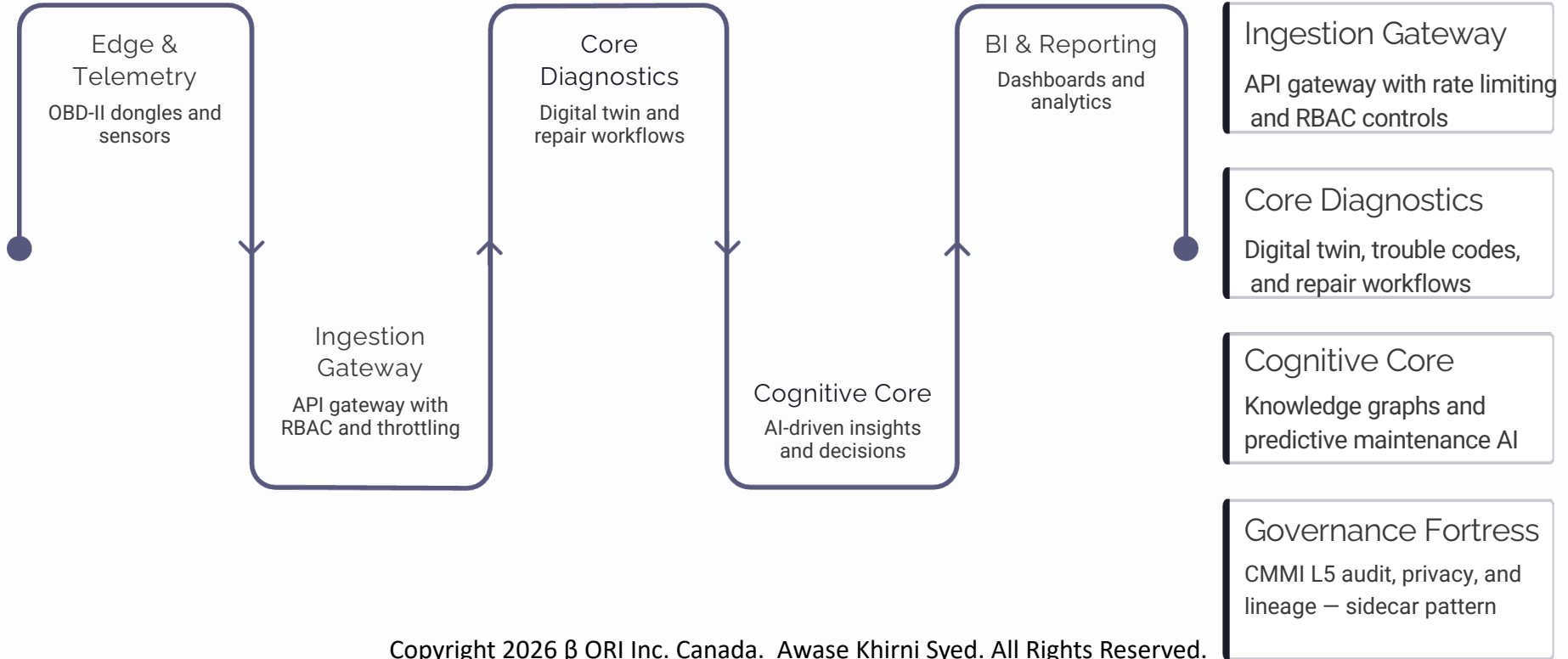
Cryptographic audit trails and a CMMI Level 5 data fortress—the trust infrastructure enterprise fleets and insurers demand before signing.

The Data Flywheel

Every connected vehicle makes the platform **smarter and more defensible**. The moat widens automatically with every new shop, every new mile of telemetry.

The Seven-Block System

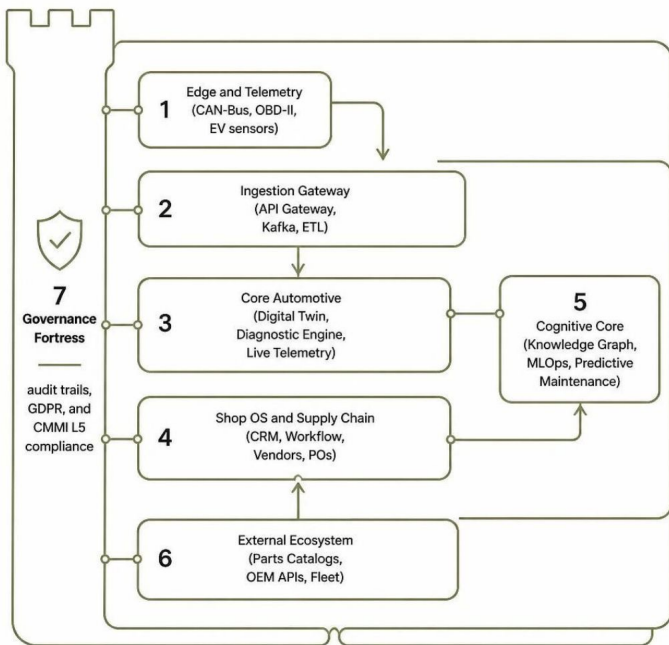
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7-Block Architecture

A unified technical platform spanning edge telemetry through cognitive intelligence — governed end-to-end by cryptographic compliance controls.



Block 1 — Edge & Telemetry

Vehicle CAN-Bus, OBD-II dongles, and EV sensors stream real-time data into the platform boundary.

Block 2 — Ingestion Gateway

API Gateway enforces RBAC and rate limits; Kafka pipeline and batch ETL normalize all inbound feeds.

Block 3 — Core Automotive

Digital Twin (Vehicles, Subsystems, Gateways), Diagnostic Engine (DTCs, Procedures, Sessions), and Live Telemetry logs.

Block 4 — Shop OS & Supply Chain

Unified shop CRM, workflow orchestration, and enterprise supply chain (vendors, POs, warehouses).

Block 5 — Cognitive Core

Diagnostic Knowledge Graph, MLOps pipelines, and Predictive Maintenance models enable proactive failure detection.

Block 6 — External Ecosystem

Parts catalogs, payment gateways, OEM APIs, and fleet integrations close the end-to-end supply chain loop.

Block 7 — Governance Fortress

Cryptographic audit trails, GDPR controls, data quality validation, and lineage tracking enforce CMMI L5 compliance across all blocks.

 **Sidecar Pattern:** Block 7 (Governance) wraps every other block — compliance is enforced at the infrastructure layer, not the application layer.



Scope

Market Opportunity

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\$500B+

Total Addressable Market
Global Automotive Aftermarket

\$85B

Serviceable Addressable Market
Independent Repair & Fleet Maintenance
Software/Services - North America & Europe

\$2.5B

5-Year SOM Target
SaaS + Hardware + Supply Chain Take-Rate

12.6

Average Vehicle Age (Years)
An all-time high - driving relentless repair demand
and EV aftermarket "Blue Ocean" opportunity



Scope

Product Experience

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Predict & Alert



Diagnose & Estimate



Fulfill & Pay



Source & Optimize



Business Model

SaaS Subscription

Tiered by bay count and shop volume

\$50–\$199 / month per location

Hardware-as-a-Service

Proprietary OBD-II Telemetry Dongles - B2B2C or B2B Fleet

\$10 / month per connected vehicle

Supply Chain Marketplace

3–5% take-rate on \$150B+ annual parts spend - near-zero marginal cost

The margin multiplier.



3 Revenue Pillars



Go-To-Market Strategy

1

Phase 1: Land

Months 1–12

High-volume, tech-forward independent shops and regional chains. "EV-Ready" messaging cuts through legacy noise.

2

Phase 2: Expand

Months 12–24

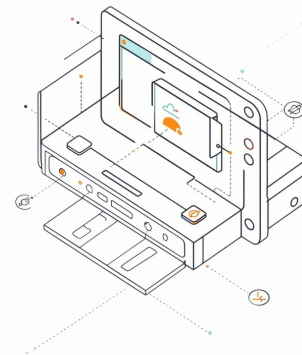
API integrations with major parts distributors. Launch B2B2C Consumer App – vehicle owners monitor their own Digital Twin health.

3

Phase 3: Dominate

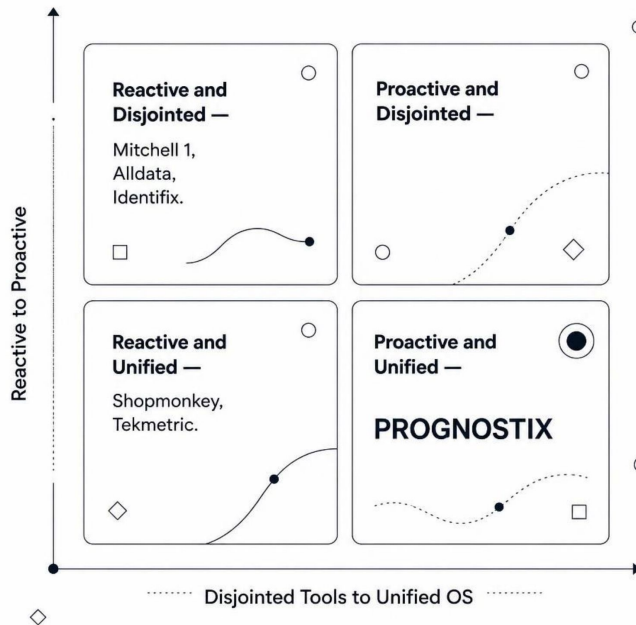
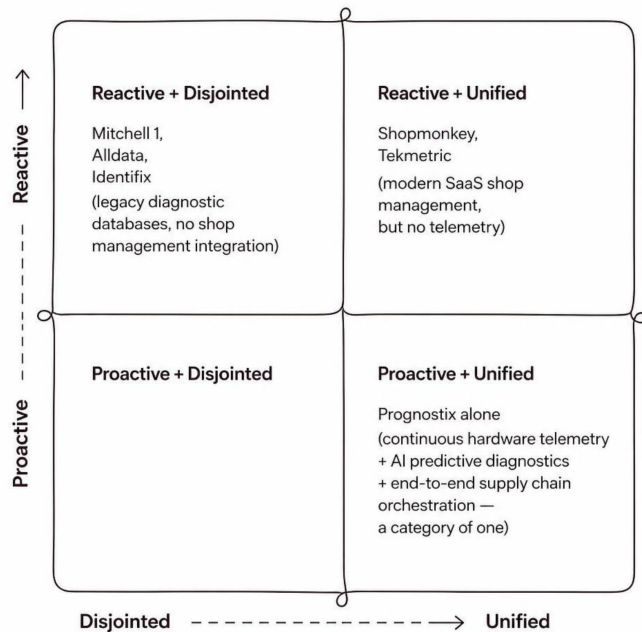
Month 24+

Enterprise Fleets: rental cars, delivery logistics. Compliance checks, driver behavior analytics, and automated predictive maintenance at scale.





Competitive Landscape



Legacy Diagnostics — Mitchell 1, Alldata, Identifix — are reactive and disjointed. Powerful databases, but no shop management integration and zero predictive capability.

Modern Shop Management — Shopmonkey, Tekmetric — unified SaaS, but entirely reactive. No telemetry. No prediction. No hardware moat.

Prognostix is the only player combining continuous hardware telemetry, AI predictive diagnostics, and end-to-end supply chain orchestration. No competitor occupies our quadrant.



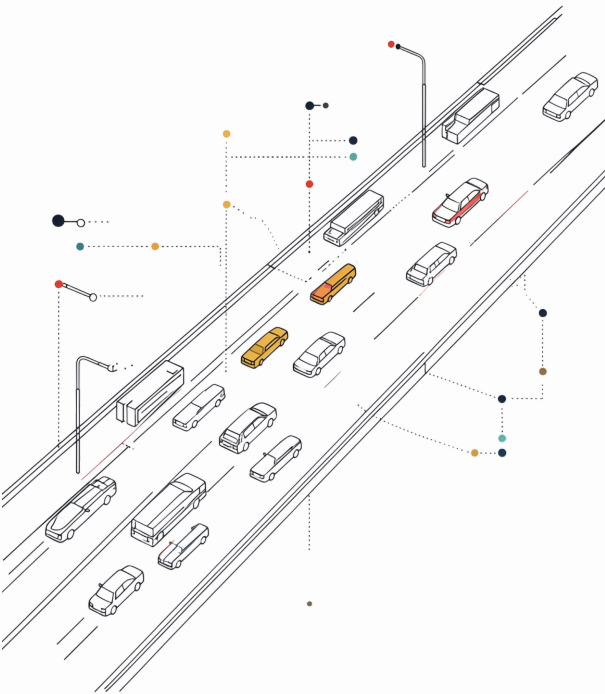
Competitive Position: Category of One

Capability	Legacy Diagnosti cs	Modern Shop Mgmt	Prognos tix
Continuous Telemetry	X	X	✓
AI Predictive Diagnostics	X	X	✓
Unified Shop Management	X	✓	✓
Supply Chain Orchestration	X	X	✓
EV Battery Analytics	X	X	✓
Court-Admissible Audit Trail	X	X	✓

Mitchell 1, Alldata, Identifix: reactive, disjointed.

Shopmonkey, Tekmetric: unified SaaS but entirely reactive.

Prognostix owns the central nervous system.



Enterprise Scale

Handles millions of vehicles, billions of telemetry events, and complex multi-vendor supply chains without compromise

Compliance First

CMMI L5 governance delivers regulatory confidence and the customer trust that accelerates enterprise deals

Competitive Moat

Proprietary knowledge graph + predictive AI shift the entire industry from reactive repair to intelligent prevention



Scope

Real-Time Data Flow

Edge to Intelligence

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Streaming Pipeline

Vehicle telemetry flows through Kafka event bus to live dashboards and predictive models in milliseconds



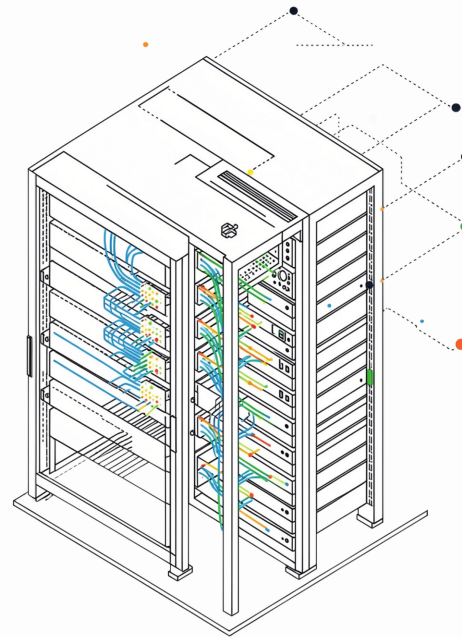
Batch ETL

OEM service bulletins and parts catalogs sync nightly, enriching the diagnostic knowledge base



Feedback Loop

AI-generated maintenance alerts flow back to Shop OS, triggering proactive parts ordering automatically



The diagram illustrates a multi-layered system architecture for a vehicle fleet management system, organized into three main layers:

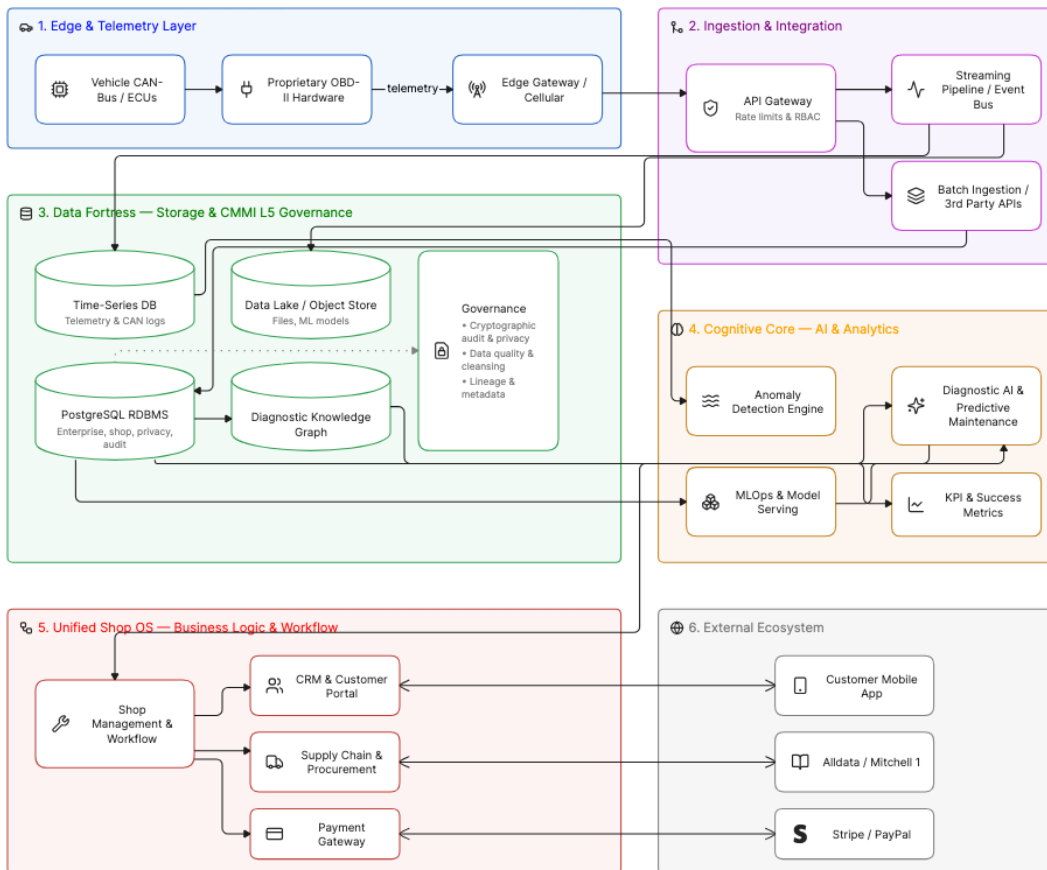
- 1. Edge & Telemetry Layer (Blue):**
 - Vehicle CAN-Bus / ECUs:** The primary data source, sending **Parts Catalog** data to the Proprietary OBD-II Telemetry Dongle.
 - Proprietary OBD-II Telemetry Dongle:** Receives data from the ECUs and sends **Continuous Telemetry CAN/PID/EV Data** to the Edge Gateway.
 - Edge Gateway (Cellular / IoT):** Receives data from the OBD-II dongle and transmits **Fleet Telemetry** data to the Ingestion Layer.
- 2. Ingestion & Integration Layer (Orange):**
 - API Gateway (Rate Limits & RBAC):** Receives data from the Edge Gateway and the Parts Catalog, and routes it to the Streaming Pipeline.
 - Streaming Pipeline (Event Bus / Kafka):** Receives data from the API Gateway and the Edge Gateway, and routes it to the Batch Ingestion.
 - Batch Ingestion (3rd Party APIs):** Receives data from the Streaming Pipeline and routes it to the External Ecosystem.
- 3. External Ecosystem (Grey):**
 - Parts Distributors (Alldata / Mitchell 1):** Receives data from the API Gateway and the Parts Catalog.
 - Customer Mobile App:** Receives data from the API Gateway and the Parts Catalog.
 - Payment Gateways (Stripe / PayPal):** Receives **Transactions** data from the Batch Ingestion.
 - Enterprise Fleets & Insurance APIs:** Receives **Fleet Telemetry** data from the Edge Gateway and **Transactions** data from the Batch Ingestion.

Key data flows include:

- Parts Catalog:** From Vehicle CAN-Bus / ECUs to Parts Distributors, Customer Mobile App, and API Gateway.
- Continuous Telemetry CAN/PID/EV Data:** From Proprietary OBD-II Telemetry Dongle to Edge Gateway.
- Fleet Telemetry:** From Edge Gateway to Batch Ingestion and Enterprise Fleets & Insurance APIs.
- Transactions:** From Batch Ingestion to Payment Gateways and Enterprise Fleets & Insurance APIs.



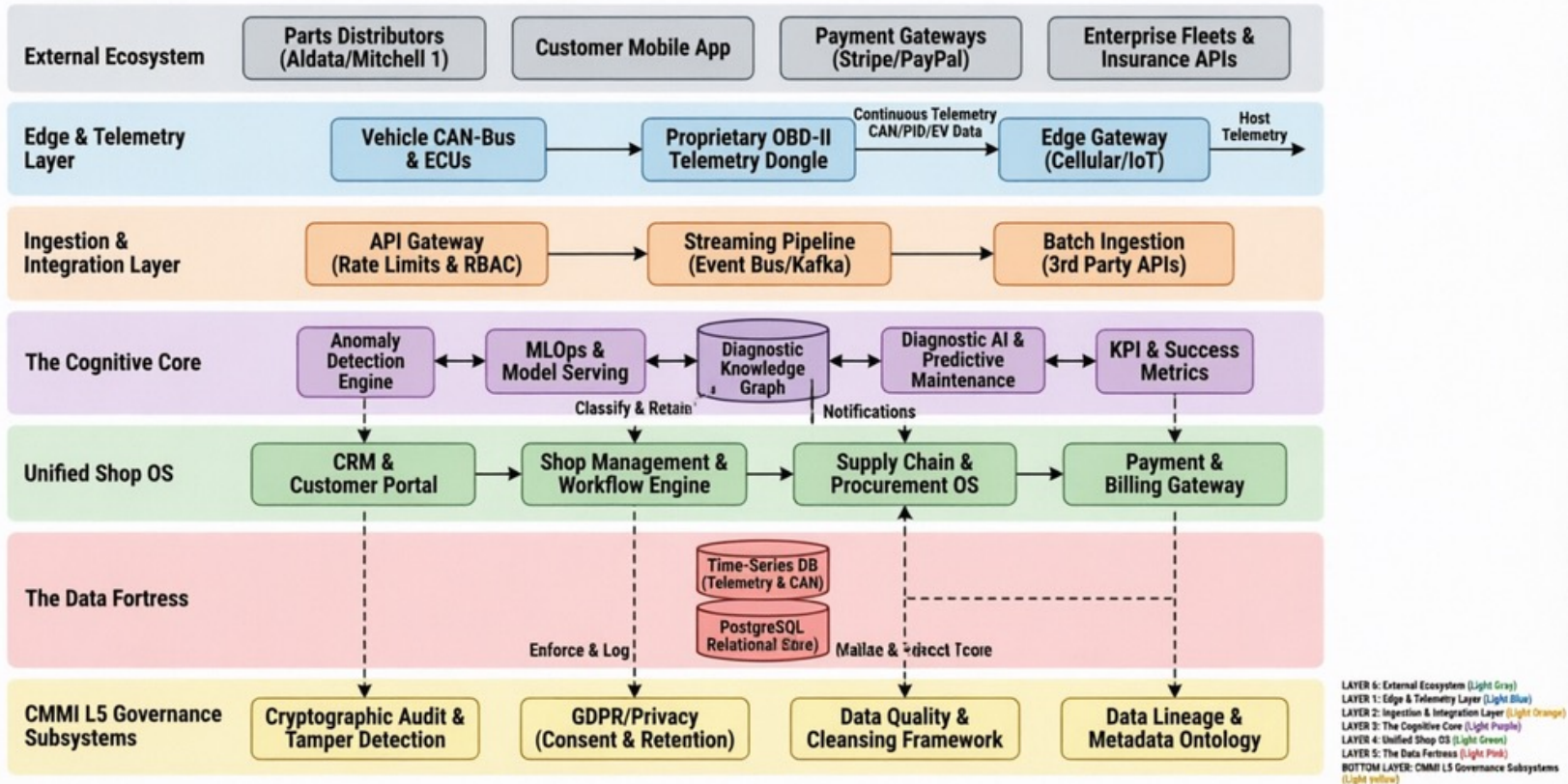
Connected Vehicle Diagnostics Platform

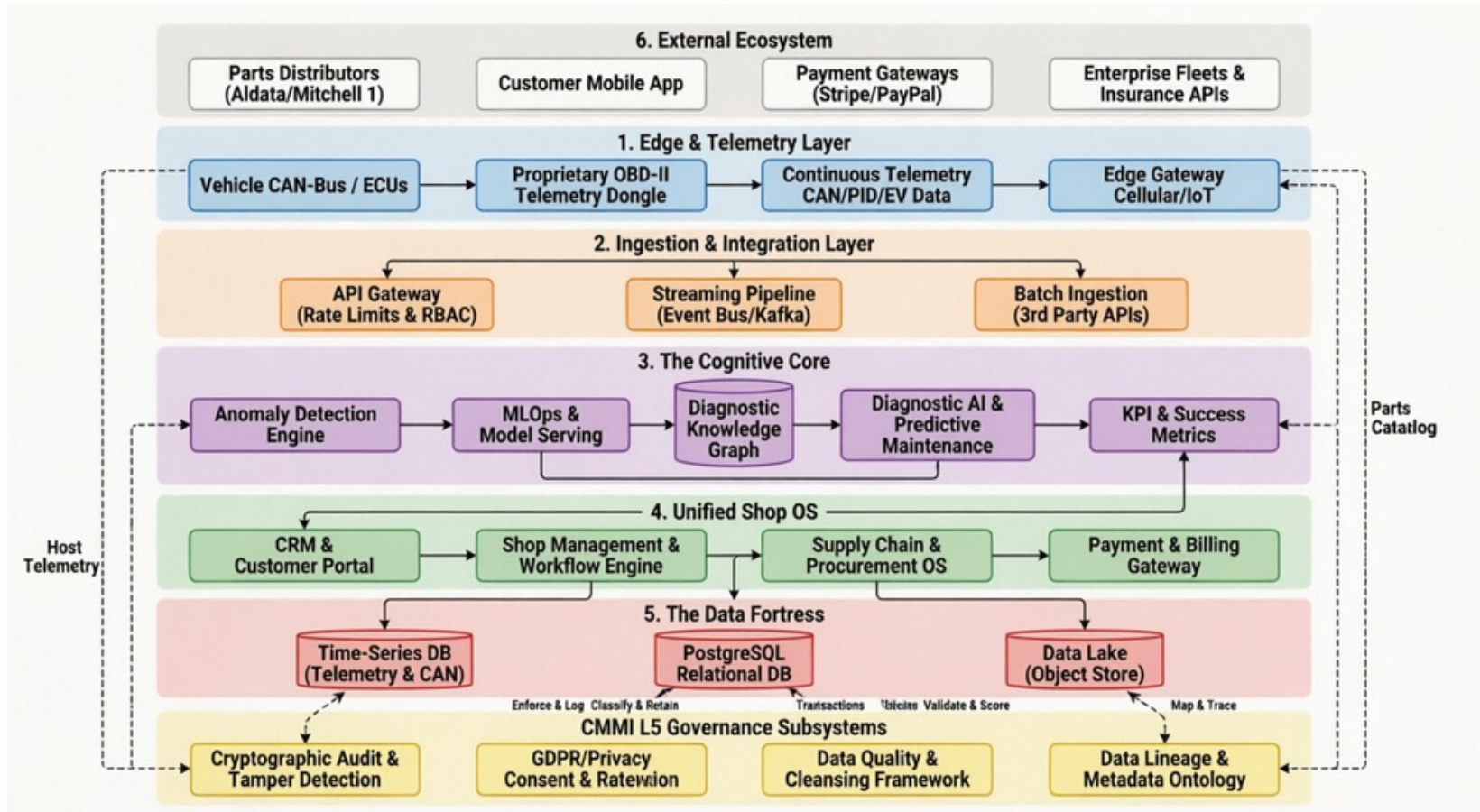




Conceptual Architecture Diagram

Edge to Intelligence







Compliance by Design



Cryptographic Audit

SHA-256 hashing creates immutable, tamper-proof logs of every database mutation



Privacy & GDPR

Automatic PII classification, consent tracking, and cross-border data transfer controls



Data Quality

Continuous validation with automated cleansing workflows for completeness and accuracy



Data Lineage

Recursive impact analysis traces data from OBD-II edge through ETL to final BI dashboards



1

Knowledge Graph

Maps non-linear relationships between symptoms, diagnostic trouble codes, and vehicle subsystems

2

ML Models

Predict component failures
30–90 days in advance
using historical telemetry and repair patterns

3

Supply Chain Orchestration

Automatically stages parts at service centers before failures occur

30–90

Days Early

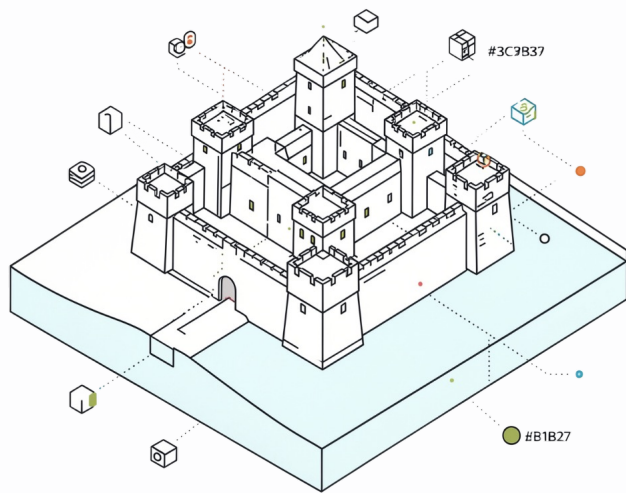
Failure prediction window

40%+

Downtime Reduction

Through proactive parts staging

Defensible Moat



Data Flywheel

Every repair trains our AI — exponentially improving accuracy the more shops join the network.



Enterprise Architecture

CMMI Level 5 compliance and cryptographic audit trails — meeting the standards large fleets demand.



EV Readiness

Native battery degradation diagnostics — years ahead of every legacy tool in the market today.



Vision & Roadmap

1

Today

AI Diagnostics, Unified Shop OS, ICE & Hybrid Telemetry — live and generating data.

2

Next 18 Months

Deep EV Battery Analytics, Automated Supply Chain Procurement, Consumer Mobile App launch.

3

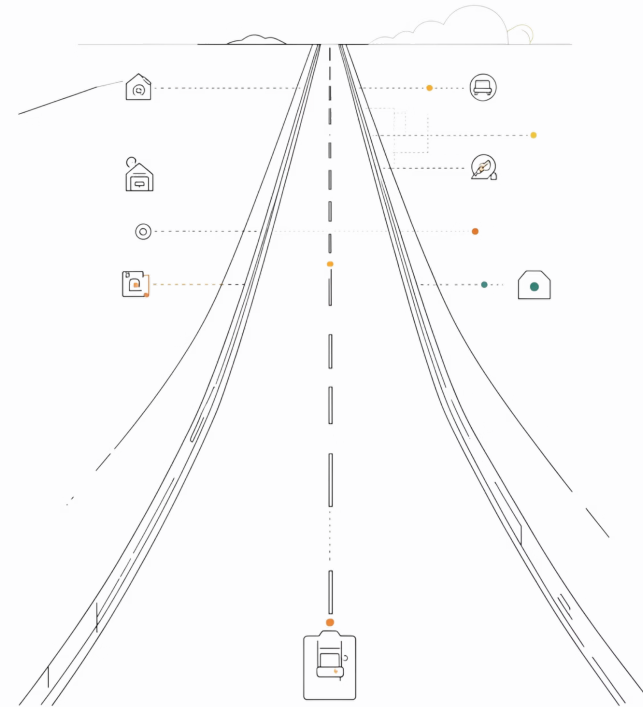
3-5 Years

Software-Defined Vehicle (SDV) Integration — OTA update verification, gateway security access, autonomous fleet health monitoring.

4

Ultimate Moat

The world's largest, most accurate knowledge graph of vehicle degradation and repair outcomes Built to last.





Scope

Unit Economics

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\$1,200 / year

ARPA – Average Revenue Per Account, per shop annually

8 Months

CAC Payback Period – among the best in vertical SaaS

4.75 : 1

LTV:CAC Ratio – driven by exceptional retention and low churn





Scope

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Year 3 Financial Vision

\$45M

Total Revenue

SaaS \$18M · HaaS \$18M · Marketplace \$9M

81%

Gross Margin

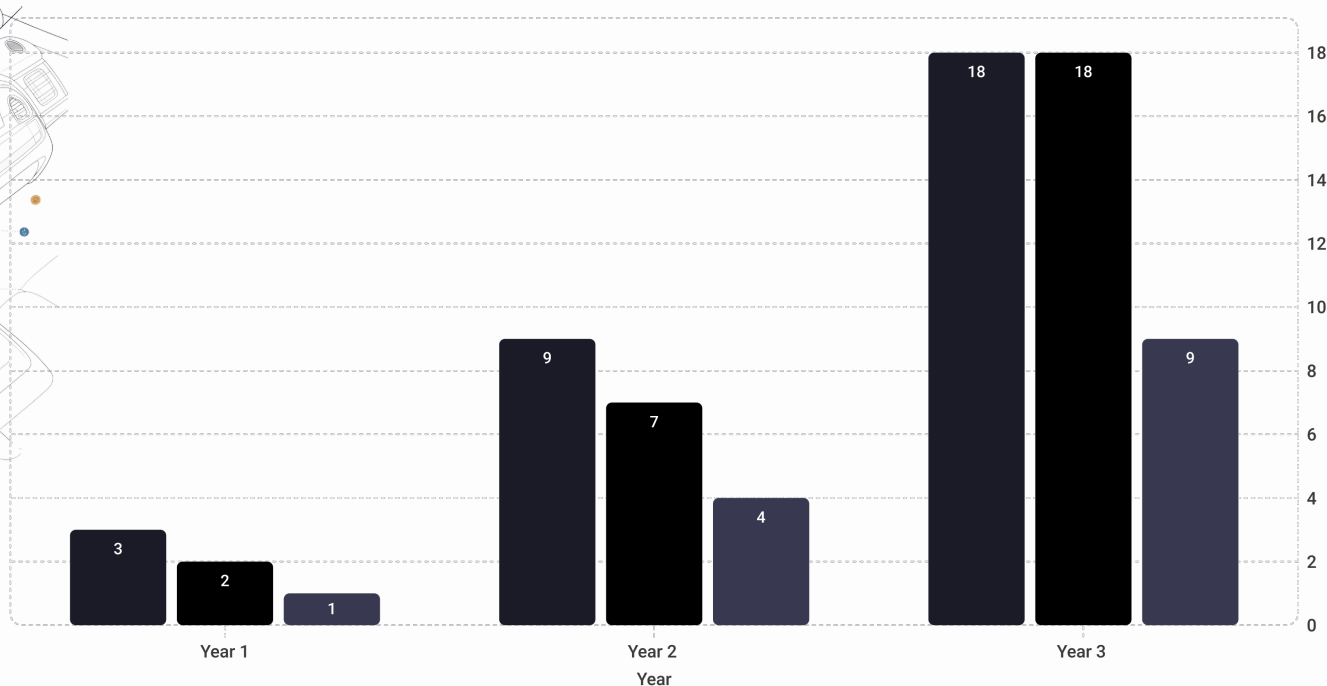
Up from 62% in Year 1 as platform leverage compounds

\$23M

EBITDA

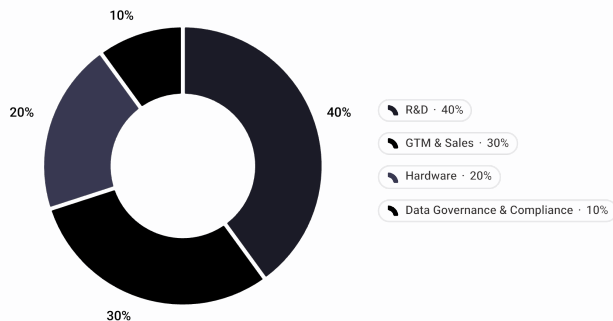
Profitable by Year 2 — exceptional capital efficiency

■ SaaS (\$M) ■ HaaS (\$M) ■ Marketplace (\$M)





The Ask & Use of Funds



40% R&D

AI model training, EV telemetry expansion, SDV gateway security

30% GTM & Sales

Scaling direct sales team and supply chain partner onboarding

20% Hardware

Integration with Blue Driver Bluetooth service to capture all telemetry data in real-time

10% Data Governance

GDPR, SOC2, CMMI L5 certifications



18-Month Milestones: 1,000 Shops Onboarded | 100,000 Vehicles Connected | \$X Million ARR



Team & Advisors: Execution Expertise

Data is only powerful when it works together. At β ORI Inc., we build end-to-end operating systems that unify spatial and aspatial data to solve complex business challenges. Unlike traditional consultancies. Our products are engineered by technology researchers who turn fragmented information into intelligent, real-time decision engines. From Strategy to execution. We don't just advise; we build the infrastructure that makes your business smarter, faster, and truly location-aware. Beta ORI: where deep science meets real-world impact.



Awase Khirni Syed Ph.D.

Founder & CEO/CTO

- 19 years experience in Academia and industry.
Technology Coach
- Strong government relations and business development acumen.
- Critical for: Strategic vision, market penetration, partnerships, Implementation, Operational Excellence, Regulatory navigation and implementation, Technology Stack Implementation, Program Management



Automotive Diagnostic Intelligence Architecture

